CHANDU PRASAD BHAIRAPU



AI Engineer

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ABOUT ME

Master's graduate in Artificial Intelligence with 3 years of professional experience in AI engineering and software development. Specialized in building and integrating backend systems, APIs and integrations, agentic AI applications and cloud deployment on Azure and AWS, with Python, FastAPI and Docker. Most recently sole developer of an agentic knowledge-extraction system for the German energy sector, working directly with domain experts, backed by consulting experience with business stakeholders at Covestro.

EXPERIENCE

Associate Generative AI Developer - Agentic Indexing Pipeline Engineer for German Energy Sector

Kniit GmbH

November 2025 – present Muldestausee, Germany

- Sole developer owning the full knowledge-extraction pipeline end-to-end – data ingestion, chunking, prompt engineering, and indexing (agentic, deterministic, and monolithic strategies) – reporting directly to the CTO and working closely with SAP domain experts across the German energy sector to turn their requirements into working solutions.
- Developed Expert Interface, a full-stack graph curation web app (React frontend, Python microservices backend) deployed on Azure, enabling domain experts to validate and approve knowledge-graph changes.
- Designed and implemented a Model Context Protocol (MCP) query service exposing the knowledge graph through an API for millisecond-latency retrieval, integrated directly into Claude Code for agentic development workflows.
- Designed and optimized multi-agent extraction algorithms (LangGraph: entity-extraction, relationship-analysis, consolidation, and reflection agents running in parallel) with iterative LLM prompting for knowledge extraction from complex regulatory and technical documents, driving precision/recall from 33%/45% to 93%/90% via an SAP-system-derived ground truth.
- Engineered a deterministic (code-based) extraction pipeline for a second, high-volume source stream, cutting a $200,000 projected LLM budget to near-zero recurring cost while pushing extraction accuracy toward 100%.
- Delivered as an AI-augmented engineer – leveraging Claude Code, custom skills, and agentic coding workflows for rapid, high-quality development grounded in clean code principles.
- **Tech stack: Python, FastAPI, REST APIs, MCP, Azure, Docker, Kubernetes, React, Postgres, MongoDB, Neo4j, LangGraph, LangChain, LiteLLM, GraphRAG, RAG**

GenAI Developer (AI Consulting Intern) - Development and Deployment of an Agentic AI Workflow

Covestro Deutschland AG

December 2024 – April 2025 Leverkusen, Germany

- Designed and deployed an agentic AI application using LangGraph to optimize decision-making workflows for vector database queries, general inference, and summarization tasks.
- Developed a React-based front-end and a Python backend with FastAPI, leveraging Postman for API testing.
- Built and containerized the entire stack with Docker and deployed it to AWS using CDK, adhering to clean code practices. Version-controlled via GitLab, with CI/CD integration.
- **Tech stack: Python(Backend), FastAPI, REST APIs, Postman, Docker, AWS, CDK, EC2, Lambda, GitLab, GitLab CI/CD, React(Frontend), LangGraph, LangChain**

GenAI Developer (Master's Thesis) - Evaluating the Performance of RAG Systems

Covestro Deutschland AG

May 2024 – November 2024 Leverkusen, Germany

- Automated RAG's Generation and Retrieval evaluation with RAGAS and PytrecEval, comparing model answers to chemist-provided ground truth.
- Built AWS cloud infrastructure with Lambda, SQS, and DynamoDB for PrismaChat RAG evaluation using CDK.
- Benchmarked multilingual embedding models on over 1M data chunks of Covestro's internal chemical data.

- Designed RAG pipelines, focusing on chunking strategies, vector databases, indexing, and fine-tuning models for optimized retrieval.
- **Tech stack: Python, Scikit-Learn, PyTorch, TensorFlow, LangChain, Hugging Face Transformers, RAGAS, PytrecEval, AWS (S3, EC2, VPC, Lambda, SQS, DynamoDB, CDK), Azure AI Search, Docker, Streamlit, Vector Databases**

Data Scientist (Intern) – ML-Based Prediction of Prepolymer Synthesis Outcomes

Covestro Deutschland AG

November 2023 – April 2024 | Leverkusen, Germany

- Built and compared Random Forest, XGBoost, and LSTM models to predict prepolymer synthesis outcomes (e.g. viscosity, reaction progression) from time-series process data, evaluating tree-based vs. deep learning approaches for this tabular/temporal hybrid problem.
- Engineered lag features and rolling statistical windows from raw sensor time-series (temperature, reaction time, process variables) to feed the tree-based models, while feeding raw sequences directly into the LSTM.
- Benchmarked model performance against traditional empirical curve-fitting methods, improving early identification of batches likely to deviate from target specification.
- Deployed an interactive Streamlit dashboard exposing model predictions to chemists in real time, letting them simulate parameter changes and see predicted outcomes before running physical batches.
- **Tech stack: Python, Scikit-Learn, XGBoost, TensorFlow/Keras, Pandas, NumPy, SciPy, Matplotlib, Streamlit**

Machine Learning Engineer - Electrical Vehicle Load Forecasting using Time Series Analysis

Cognizant Technology Solutions

September 2021 – March 2022 | Hyderabad, India

- Developed ARIMA, SARIMA, XG-Boost regressor, FB Prophet, LSTM models for time series analysis for EVs.
- Developed LSTM, achieved 96% test accuracy and strong performance on unseen data.
- **Tech stack: Python, Scikit-Learn, TensorFlow**

PROJECTS

Project: Lab Requests Portal | TypeScript | React | PostgreSQL | Azure

- Internal web application where teams file and track requests: React + TypeScript frontend, TypeScript (Node.js) REST API, PostgreSQL schema with migrations and row-level security, single sign-on with Entra ID (OIDC) and role-based access control, and a scheduled job that syncs data from an external system.
- Automated tests, structured logging, request metrics and OpenTelemetry traces, CI/CD in GitHub Actions deploying to Azure Container Apps, and handover documentation and runbooks for the long-term owner.

TECHNICAL SKILLSET

- **Programming:** Python, SQL, React(beginner)
- **Programming Tools/Frameworks:** FastAPI, REST APIs, Postman, Docker, Kubernetes, Agentic Workflow(LangGraph), LangChain, NLP (Hugging Face transformers, NLTK), Machine Learning (Scikit-Learn), Deep Learning (PyTorch, TensorFlow), Protege
- **Databases:** SQL, Postgres, SPARQL, Vector Databases(Qdrant, Pinecone, Azure AI Search)
- **Cloud technologies:** Azure, AWS- S3, EC2, Lambda, SQS, Load balancer, DynamoDB, VPC, CDK
- **Version Control & Methodologies:** Git, Agile

EDUCATION

Master of Science in Artificial Intelligence — March 2022 – April 2025
Technische Hochschule Würzburg-Schweinfurt, Würzburg, Germany

Minor in Computer Science & Engineering — July 2019 – July 2021
G Pulla Reddy Engineering College, Kurnool, India

Bachelor of Technology in Mechanical Engineering — August 2017 – July 2021
G Pulla Reddy Engineering College, Kurnool, India

LANGUAGES

German (B1) | English (Business Fluent) | Hindi (Fluent) | Telugu (Native)